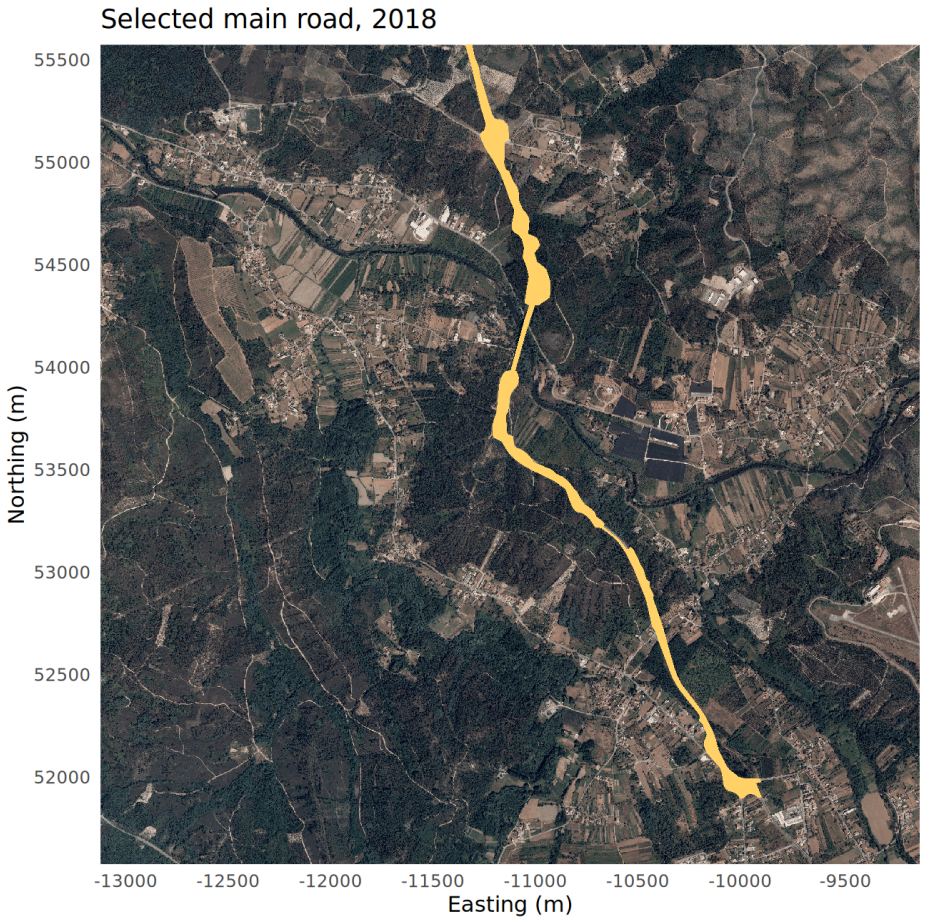
BIOL6130 Final Project: eucalyptus near roads

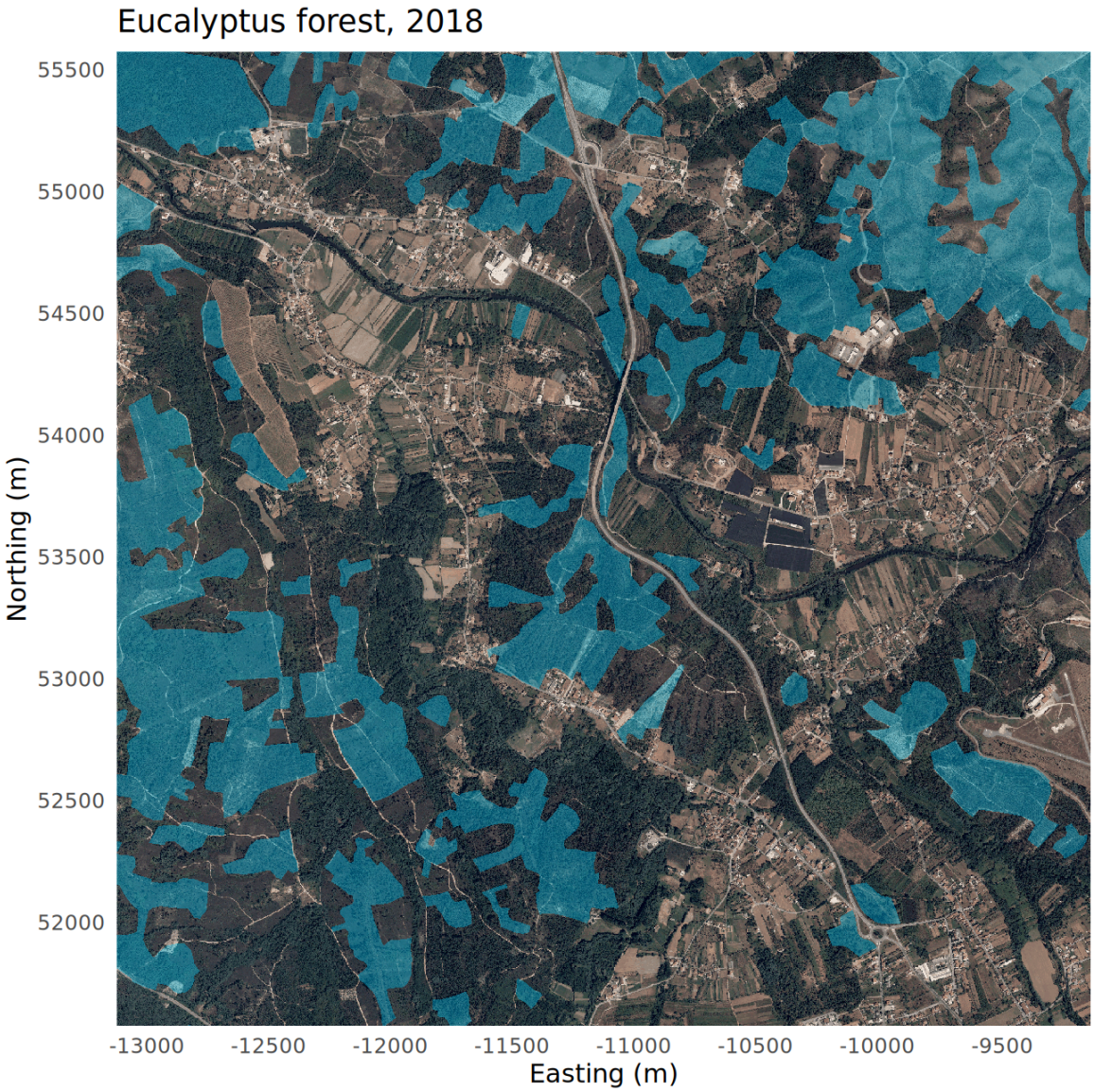
Pilot study in area B

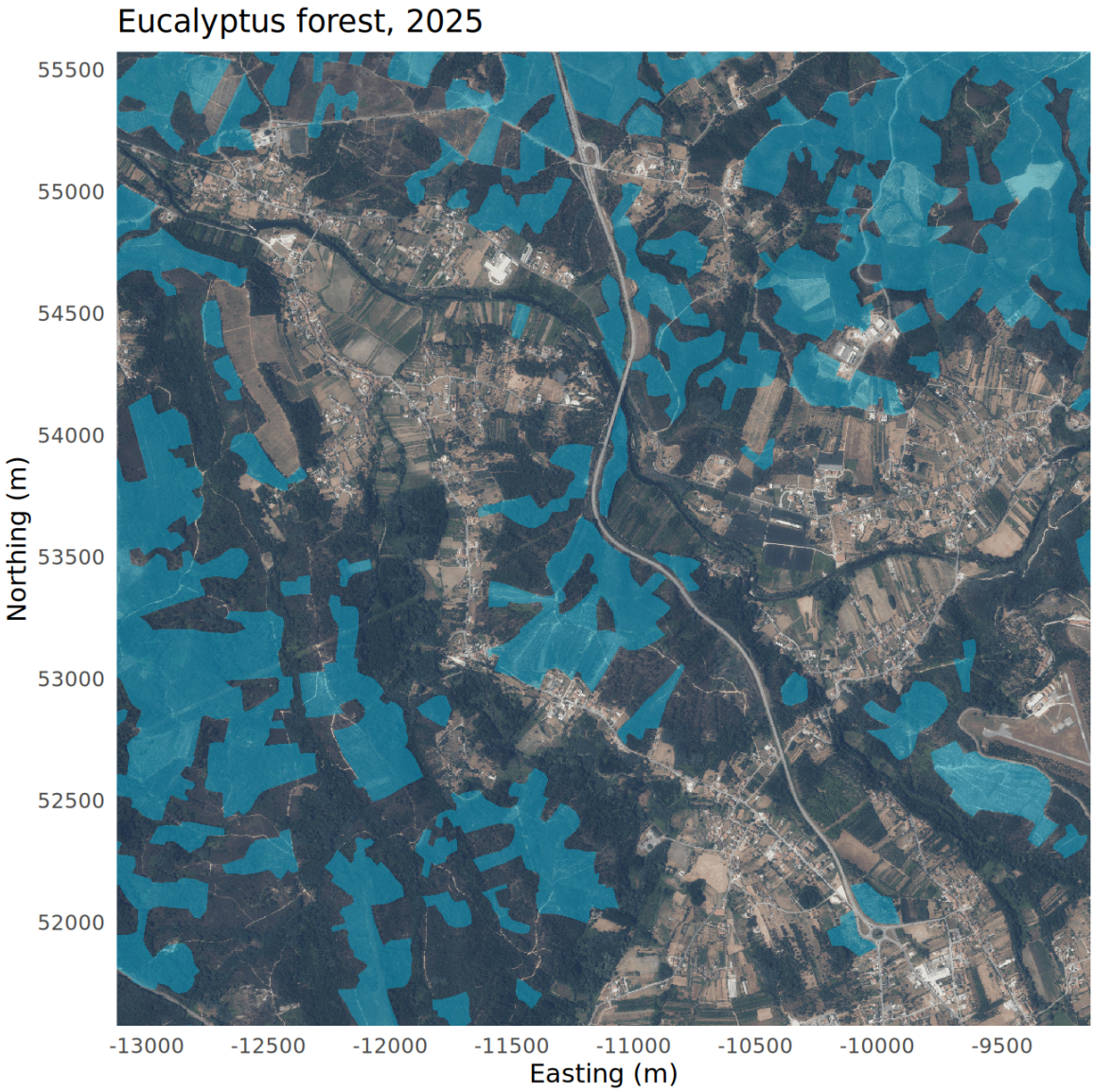
We first looked at a 4 km by 4 km area near Coimbra, Portugal. The yellow line on the first map shows the main road we selected.

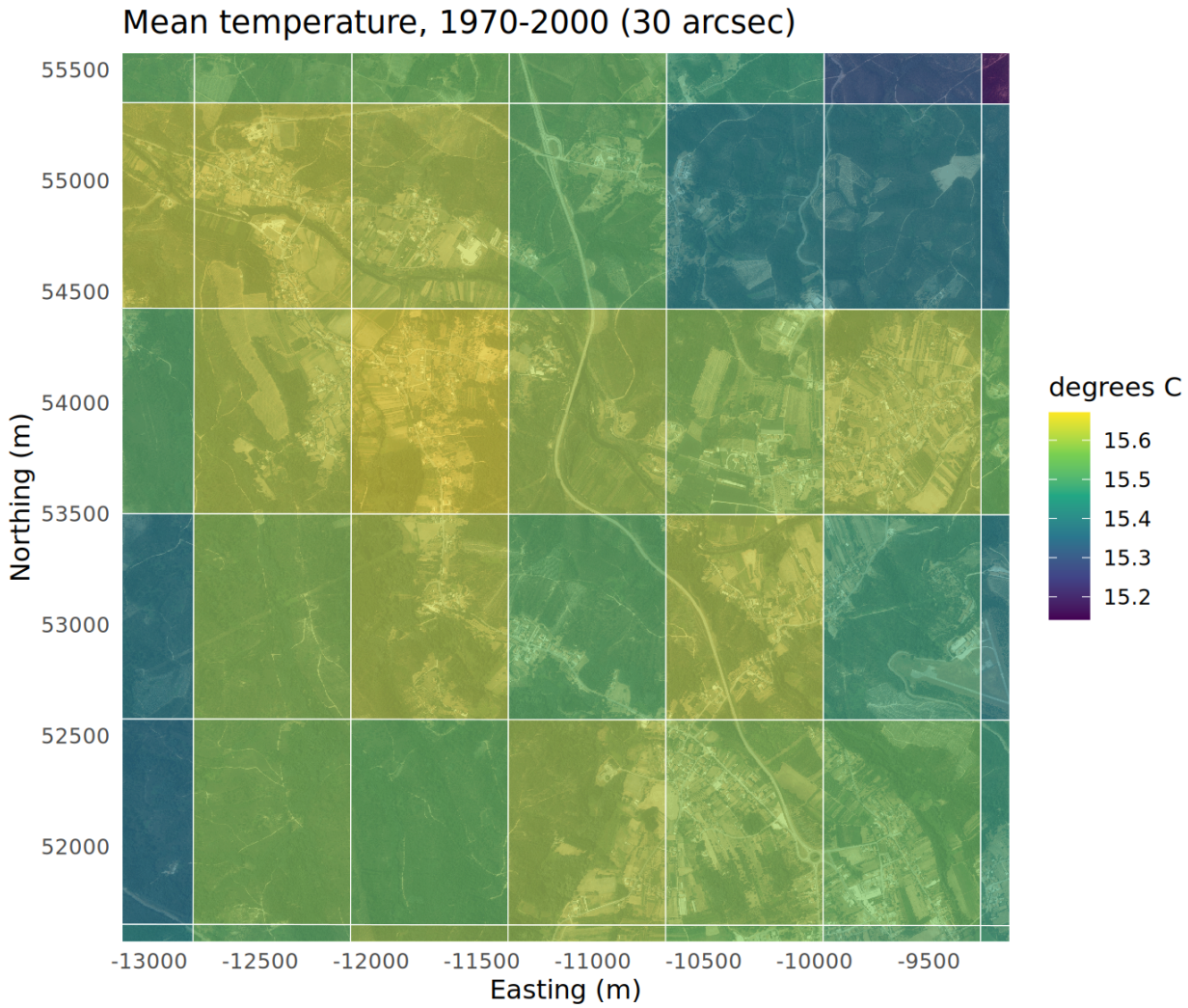
Four fields from the 2018 land-cover map are shown below.

```
Rows: 6,063
Columns: 4
$ id         <int> 1288, 1294, 1296, 1297, 1299, 1323, 1361, 1367,~
$ cover_code <chr> "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.~
$ cover_name <chr> "Áreas edificadas residenciais contínuas predom~
$ area_ha    <dbl> 2.5112068, 4.9795026, 1.3266062, 7.0387354, 2.8~
```

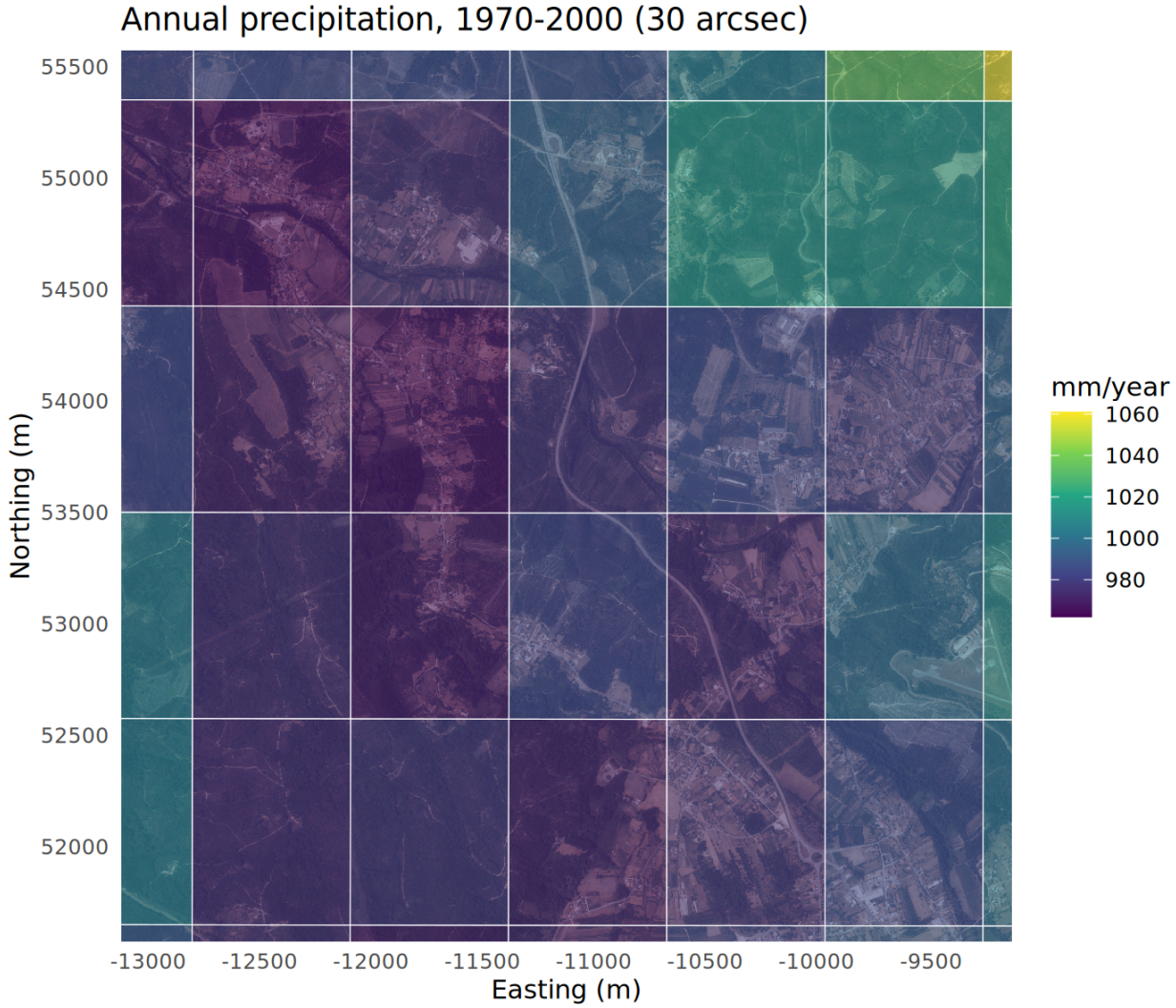



Eucalyptus forest, 2018
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)

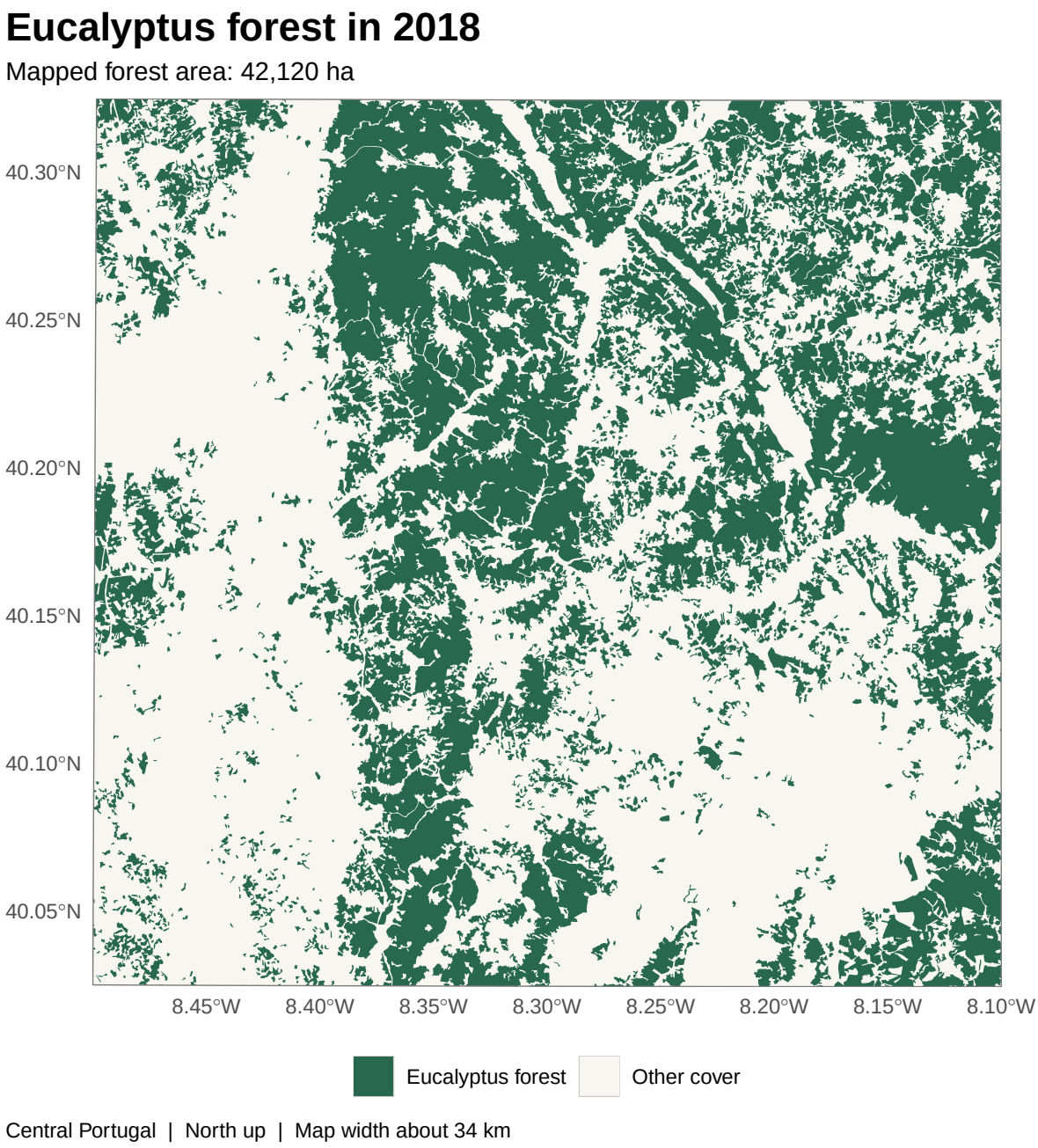

Eucalyptus forest, 2025
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)



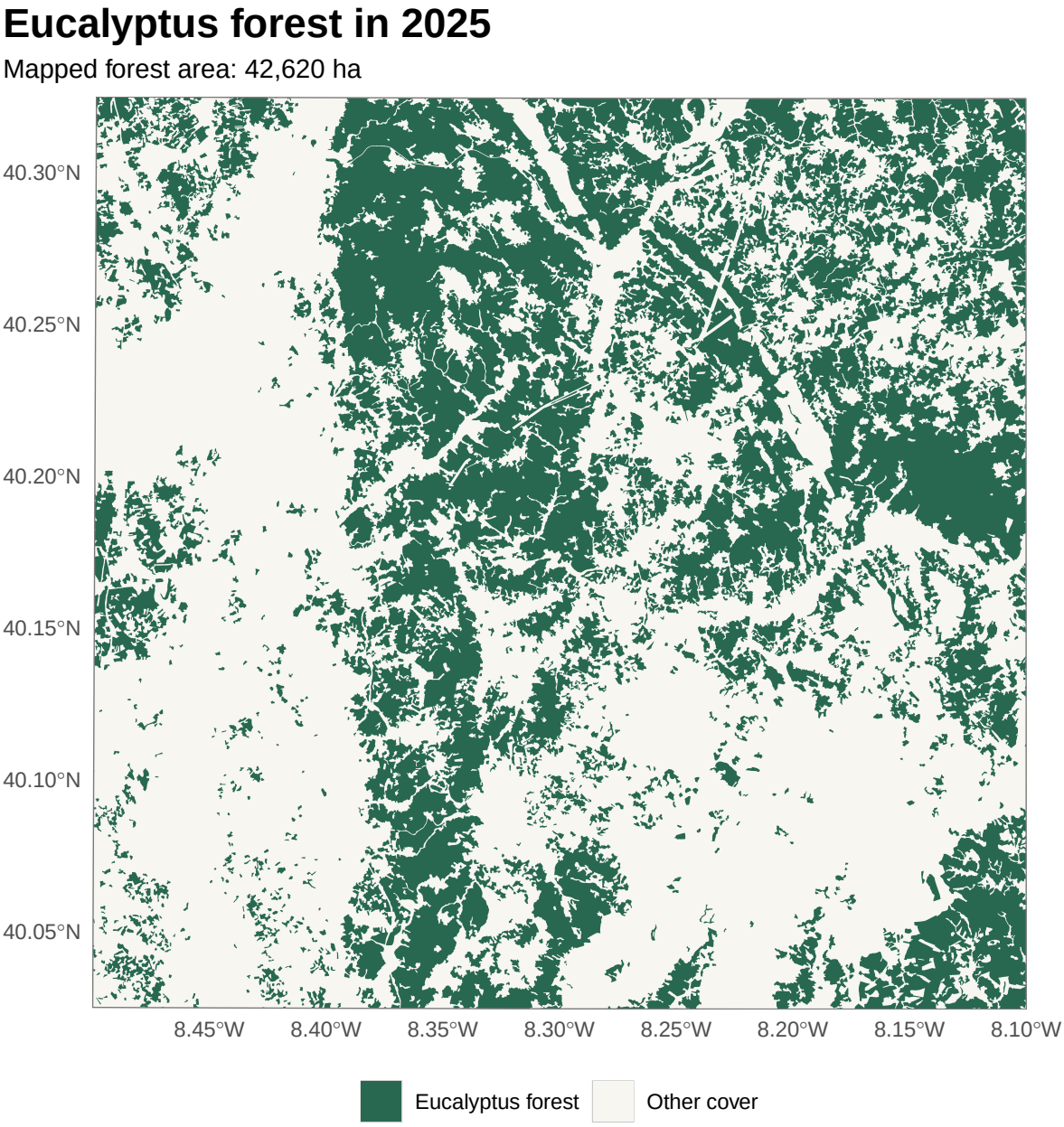
The climate maps show average temperature and annual rainfall for 1970–2000 over the 2025 aerial photo. Each climate cell covers about 710 by 925 m here.


Annual precipitation, 1970-2000 (30 arcsec)
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)
mm/year
1060
1040
1020
1000
980

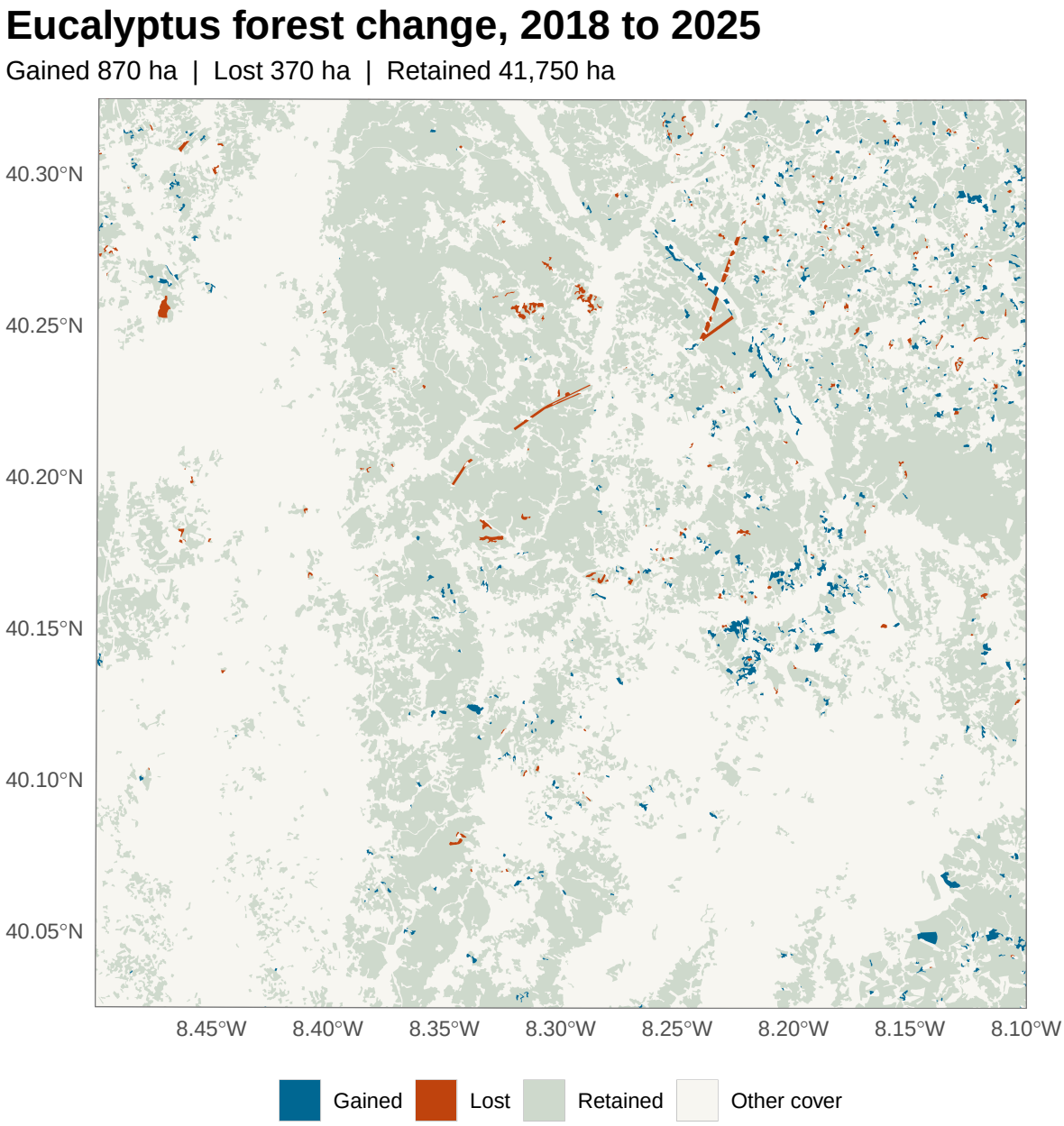
Regional eucalyptus forest cover



Green shows forest mapped as mainly eucalyptus in 2018. The study area is near Coimbra in central Portugal (8.50–8.10°W, 40.025–40.325°N).



The 2025 map shows eucalyptus forest in the same area and at the same scale.



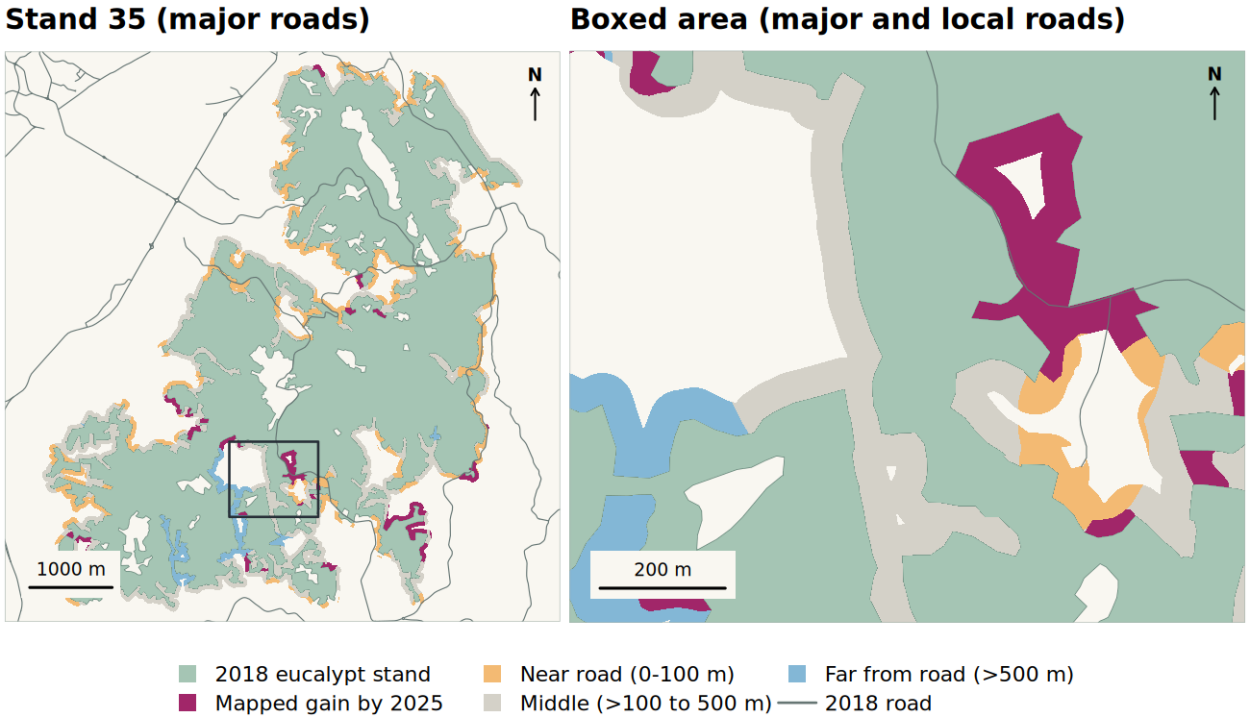
Blue shows eucalyptus forest gained between 2018 and 2025. Orange shows forest lost, and pale green shows forest mapped in both years. The net gain was **500.3 ha**. These maps show the main land cover in each area.

Environment table

Each row in `environment.csv` is one point on the 80 m grid. We used its climate, elevation, slope, land cover, fire and forest management data to find similar sites near and far from roads. Nearby points can share a climate cell.

Result 1

On the 80 m grid, 66 eucalyptus stands had sites both near a road (up to 100 m) and far from a road (over 500 m). We kept the 11 stands where the near and far sites had similar conditions. We looked at a strip of land up to 50 m outside each stand's 2018 edge and left out clearings inside the stand. Purple shows where the map changed from another land cover in 2018 to eucalyptus in 2025.



Around stand 35, the map showed new eucalyptus cover on 6.71% of the land we compared near roads and 5.60% far from roads.

Across all 11 stands, we found **55.33 ha** of new eucalyptus cover in the strips. Within these strips, **22 of the 55 new patches** reached land within 100 m of a road. The other 33 stayed farther away. We counted land shared by two stands once.

Road distance	Land compared (ha)	Gain (ha)	Share of total gain	Land with gain
Up to 100 m	763.37	14.05	25.4%	1.84%
Over 100 to 500 m	1237.94	37.42	67.6%	3.02%
Over 500 m	246.52	3.86	7.0%	1.57%

In 8 stands we could compare near and far land with similar cover, fire history, plantation status, terrain, climate and distance from the forest edge. Within each group of similar land, we gave both sides the weight of the smaller area. This gave 17.94 ha per side in total. We then gave each stand equal weight. New eucalyptus cover averaged **3.80% near roads** and **3.44% far from roads**. Only 2 stands showed any gain on this land. This gives little support for the idea that roads help eucalyptus spread. The maps cannot tell us whether the trees were planted or grew naturally.

Data sources

- **Land cover and the pilot road.** Direção-Geral do Território (DGT), COS Series 2, 2018 v4 and 2025 v1. The forest maps use eucalyptus class 5.1.1.6. DGT, CC BY 4.0
- **Regional roads.** We used the 1 January 2018 road network in `roads.gpkg` and excluded tracks. © OpenStreetMap contributors. Geofabrik Portugal archive · ODbL 1.0
- **Pilot aerial photos.** DGT imagery from 2018 and 2025, exported at 2 m for area B. DGT orthophotos, CC BY 4.0
- **Pilot climate.** WorldClim v2.1, temperature and rainfall averages for 1970–2000. WorldClim · Fick and Hijmans (2017)
- **Regional climate.** CHELSA v2.1, temperature, rainfall and mean near-surface wind speed for 1981–2010. CHELSA · BIOCLIM+ data, CC0
- **Elevation and slope.** Copernicus DEM GLO-30, tile N40/W009. Product and licence
- **Fire.** ICNF, *Áreas ardidas*, 2017–2025, retrieved 5 November 2026. Fire perimeters · ICNF terms. The 2017 burn severity and image quality layers come from Jahanianfard et al. Portuguese Burn Severity Atlas v2, CC BY 4.0
- **Forest management.** Lesiv et al., global forest management map for 2015, v3.2. Data, CC BY 4.0

Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. The organisations in charge of the Copernicus programme by law or by delegation do not incur any liability for any use of the Copernicus WorldDEM-30.